Deterministic Optimization

Linear Optimization Modeling

Handling Nonlinearity

Andy Sun
Assistant Professor
Stewart School of Industrial and Systems Engineering

Radiation Therapy

Modeling using Linear Programs

Learning Objectives for this Lesson

- Recognize basics of radiation therapy
- Identify beamlet planning problem

Overview of Radiation Therapy

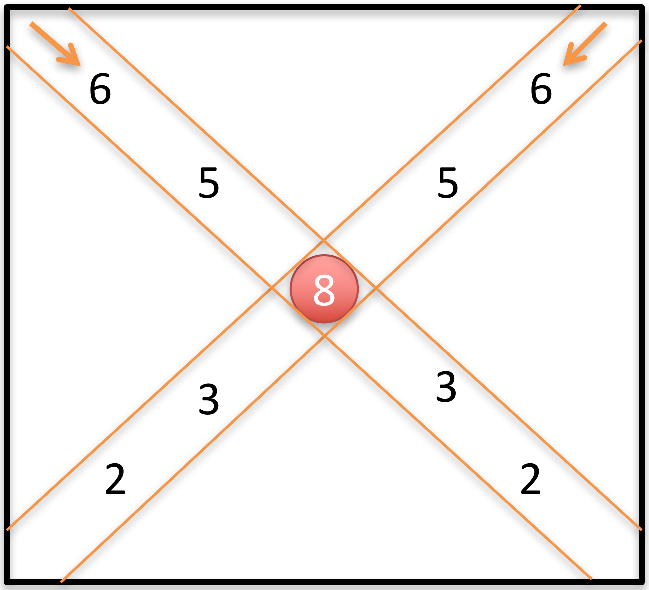
- In 2017, there will be estimated 1,668,780 new cancer cases diagnosed and 600,920 new deaths in the US (2nd most common cause of death in the US)
- Nearly 2/3 of the patients will receive radiation therapy during illness
  - Beams of radiation delivered to the body in order to kill cancer cells
- High doses of radiation can kill cells and/or prevent them from growing and dividing
  - True for cancer cells **and** normal cells
- Radiation is attractive because the repair mechanisms for cancer cells is less efficient than for normal cells

Radiation Therapy Procedure

- High accuracy 3D imaging of tumor area
  - CT: Computed tomography
  - MRI: Magnetic Resonance Imaging
- High accuracy delivery of radiation
  - IMRT: Intensity Modulated Radiation Therapy
  - Computer controlled linear accelerator
  - Hundreds of beamlets of different angles and shapes
- Need advanced Optimization models for planning the beamlets

Source: https://www.radiologyinfo.org/en/info.cfm?pg=imrt

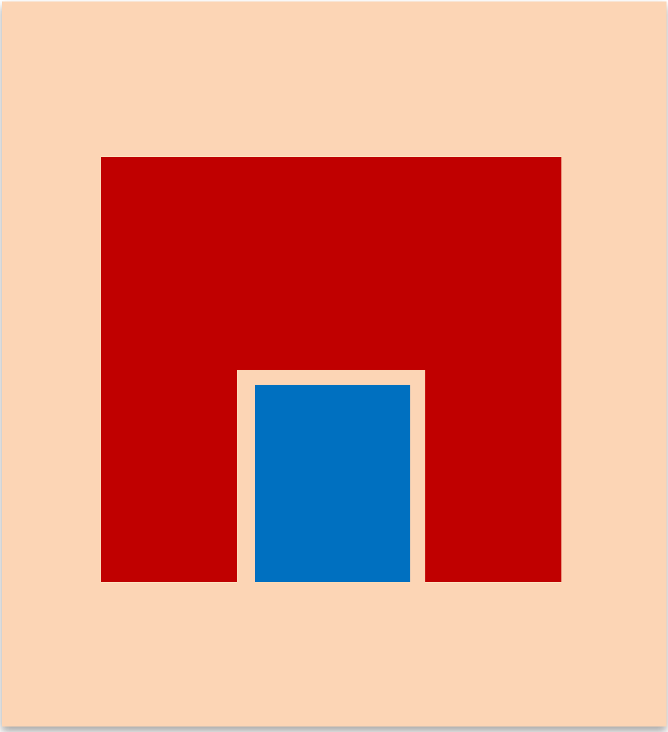
Dose Delivery



Tumor

k: Does density

Complex Shaped Tumor



Complex shaped tumor

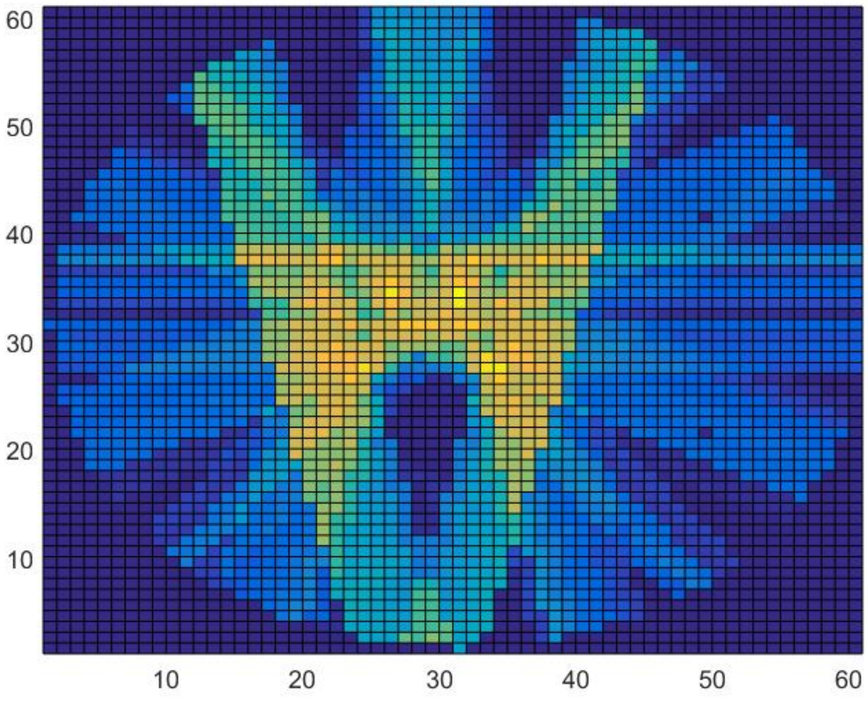
Critical area

A Radiation Therapy Planning Problem

- For a given tumor and given critical areas obtained from imaging, and a given set of possible beamlet origins and angles

- Determine the weight of each beamlet such that
  - Dosage over the tumor area will be at least a target level γ_L
  - Dosage over the critical area will be at most a target level γ_U

Radiation using Optimized Beamlets



Summary

- We introduced the background of radiation therapy.

- Set the stage for developing an optimization model for radiation therapy planning.

Conventional Radiation Therapy

- In conventional radiation therapy:
  - 3 to 7 beams of radiation
- Radiation oncologist and physicist work together to determine a set of beam angles and beam intensities
- Manual process by “trial-and-error” process
- With only a small number of beams, it is usually difficult or impossible to deliver required dose to tumor without impacting the critical area.